MIL-S-55557A(EL)
4 December 1970
Superseding
MIL-S-55557(EL)
5 June 1967

MILITARY SPECIFICATION

SHELTER, ELECTRICAL EQUIPMENT
S-330()/TRC-117(V)

1. SCOPE

1.1 Scope.- This specification covers the air or vehicular transportable shelter facility part of the radio terminal set AN/TRC-151()(V) which is capable of providing a dual multiplex terminal for 12 or 24 channels of voice frequency communications for radio or cable transmission. (See paragraphs 6.1 and 6.5).

1.2 Description.- Shelter, Electrical Equipment S-330()/TRC-117(V) hereinafter called the shelter facility, includes a modified S-280()/G shelter and all racks, brackets, shelves, tie-downs, electrical components and wiring to enable installation and operation of the assemblage components of the AN/TRC-151()(V) (See 6.1 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronic and Associated Electrical Equipment)

FSC-5820

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the current version before use.

MIL-W-6858	Welding,Resistance,Aluminum Magnesium,Non-Hardening Steels or Alloys,Nickel Alloys,Heat-Resisting Alloys and Titanium, Spot and Seam
MIL-T-7928	Terminals,Lug and Splice,Crimp Style Copper
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-W-45205	Welding,Gas,Metal-Arc & Gas,Tungsten-Arc, Aluminum Alloys,Readily Weldable,for Structures,Excluding Armor
MIL-S-55505	Shelter,Electrical Equipment(Containing Equipment) Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-252	Wired Equipment,Classification of Visual and Mechanical Defects
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-1235	Single and Multilevel Continuous Sampling Procedures & Tables for Inspection by Attributes

DRAWINGS

ELECTRONICS COMMAND

SC-A-46691	RF Cable Assembly Tests
SC-C-532288	Schematic Diagram Voltage Cutout Protector
SC-D-539634	Signal Entrance Box Assembly Wiring Diagram

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

DL-SC-B-547577	Shelter,Electrical Equipment, S-330( )/TRC-117(V)
SC-D-547753	Power Schematic
SC-D-547769	Cable Wiring Diagram (Signal Entrance Box)
SC-F-585142	Signal Schematic Wiring Diagram

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 Preproduction sample.- Unless otherwise specified in the invitation for bids and contract, the contractor shall furnish one preproduction sample of the shelter facility for approval.(See 4.3)

3.2 Construction.- The shelter facility shall be constructed in accordance with drawings as listed on DL-SC-B-547577.(See 4.6).

3.3 Electrical installation.-

3.3.1 Power system wiring and components.- Installation of power wires and components (including cable assemblies, receptacles, switches, meters, circuit breakers, neon lamps, fluorescent lamps, incandescent lamps, lighting fixtures, distribution assembly system, and wiring components) constituting a three wire electrical power distribution system, shall be installed and wired in accordance with the drawings. The complete in-place power distribution system shall meet the following requirements:

3.3.1.1 Continuity of conductors.- The point-to-point resistance of all power-line conductors shall be less than 3 ohms when tested in accordance with 4.7.1.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

3.3.1.2 Dielectric strength of insulation.- The insulation of all power-line wires, cables, and components shall withstand a potential of 500 volts DC applied continuously for not less than 30 seconds when tested in accordance with 4.7.1.2.

3.3.1.3 Insulation resistance.- The resistance of insulation between any two of the three power-line conductors and also between each of the two ungrounded power-line conductors and shelter facility ground terminal shall not be less than 100 megohms when tested in accordance with 4.7.1.3.

3.3.2 Overvoltage control.- The overvoltage control shall be fabricated and wired in accordance with the drawings and meet the following requirements:

3.3.2.1 Dielectric strength of insulation.- The dielectric strength of insulation of each overvoltage control power wire shall withstand a potential of 500 volts DC continuously applied for not less than 30 seconds when tested in accordance with 4.7.2.1.

3.3.2.2 Insulation resistance.- The resistance of insulation between each of the overvoltage control power wire shall not be less than 100 megohms when tested in accordance with 4.7.2.2.

3.3.2.3 Voltage drop.- A voltage drop of less than 250 millivolts shall exist between the respective input and output pins of the overvoltage control unit when the circuit current is 50 amps. (See 4.7.2.3).

3.3.3 Power distributional and operational functions.- Main power branch distribution and circuit switching functions thereof shall meet the operational requirements as shown by Power Schematic Drawing SC-D-547753, when tested in accordance with 4.7.3.

3.3.4 Signal (audio) system wiring and components.- Installation of signal (audio) wires, cables and wiring components (including binding posts, connectors, plugs, jacks, receptacles, and wiring devices) shall be installed and wired in accordance with the drawings. The completed in-place signal system installation shall meet the following requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the current version before use.

3.3.4.1 Continuity of conductors.- The resistance between both end terminations of all signal-line conducturs shall not exceed 5 ohms when tested in accordance with 4.7.4.1.

3.3.4.2 Dielectric breakdown voltage.- The dielectric strength of insulation of all signal wires, cables, and components shall withstand a potential of 500 volts DC applied continuously for not less than 5 seconds when tested in accordance with 4.7.4.2.

3.3.4.3 Insulation resistance.- The resistance of insulation between each signal conductor and all other conductors connected to ground shall not be less than 100 megohms when tested in accordance with 4.7.4.3.

3.3.5 Video and radio frequency (RF), system wiring and components.- Installation of video and RF cable assemblies, and connecting components (including plugs, connectors, adapters, fittings, receptacles, and lightning protectors) shall be in accordance with the drawings. The completed in-place video and RF system installation shall meet the following requirements:

3.3.5.1 Continuity of conductors.- The point-to-point resistance between the following video and RF cable terminations shall not exceed 3 ohms when tested in accordance with 4.7.5.1:

a. Both end terminations of the inner conductor.

b. Both end terminations of the outer conductor(shield).

3.3.5.2 Dielectric and insulation resistance tests.- The insulation of all video and RF cables, components and lightning protectors shall withstand the following potentials with the insulation resistance not less than 100 megohms when tested in accordance with 4.7.5.2:

a. 1500 volts DC applied continuously for not less than 30 seconds to all video cable assemblies terminated at the video entrance box with lightning protector elements removed.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the current version before use.

b. 1100 volts DC applied continuously for not less than 30 seconds to all video cable assemblies terminated at the video entrance box with lightning protector elements in place.

c. 1700 volts RMS, 25-100 cycles, applied continuously for not less than 30 seconds to all RF transmission line terminated at the antenna entrance box.

d. 1500 volts DC applied continuously for not less than 5 seconds to all video cable assemblies terminated at the video and radio patch panel.

3.3.5.3 Voltage standing wave ratio (VSWR), RF transmission lines.- The VSWR of all completed RF transmission lines shall not exceed 1.20 at transmission frequencies of 600 megahertz, 1000 megahertz, 1350 megahertz and 1850 megahertz when tested in accordance with 4.7.5.3.

3.3.6 Cable assemblies.- (see 4.4)

3.3.6.1 Cable assemblies, power and audio frequency.- All power and audio frequency cable and cord assemblies shall be fabricated in accordance with the drawings unless otherwise stated and shall meet the requirements and tests of MIL-C-3885.

3.3.6.2 Cable assemblies, radio frequency.- Radio frequency cable assemblies required for operation of major items of equipment in 6.1 shall be fabricated in accordance with the drawings unless otherwise stated and shall meet the requirements and tests as indicated in Drawing SC-A-46691.

3.3.7 Suppression of electromagnetic interference.- The motors in the ventilating equipment and heaters used in the shelter facility shall meet the emanation limits of MIL-STD-461 for tactical equipment, if other than those items defined by the drawings (see 4.4).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the current version before use.

3.4 System operation.- With all operational equipments of the AN/TRC-151() assemblage installed, the operational requirements of all power, audio and radio frequency cable assemblies, switches, connectors, and patch panels, shall be met when tested in accordance with 4.8.

3.5 Vehicular transportation.- The shelter facility shall be capable of being transported over cross-country terrain by military vehicles without sustaining any permanent damage such as buckling, cracking, delamination, or other permanent deformation.(See 4.9).

3.6 Railroad transportation.- The shelter facility shall be capable of being loaded, blocked, and braced on a flatcar for shipment by rail. When so loaded, the shelter facility shall be subjected to and withstand the test of paragraph 4.10 without sustaining any damage as mentioned in paragraph 3.5.

3.7 Drops,flat and rotational.- The shelter facility shall be capable of withstanding drops of 18 inches onto concrete. The shelter facility shall be subjected to and withstand the tests of 4.11.1 and 4.11.2 without any damage as mentioned in paragraph 3.5.

3.8 Watertightness.- The shelter facility shall be capable of withstanding the test of 4.12 without any leakage into the facility.

3.9 Fording.- The shelter facility shall be capable of being immersed in water to a depth of 21 inches, measured from the bottom of the skids, for 1 hour without any leakage of water into the facility. (See 4.13).

3.10 Mounting inserts.- All mounting inserts shall be installed without the use of any auxiliary materials such as adhesive,and when installed, each insert shall be capable of withstanding the tensile load and the torque load referenced in 4.14. There shall be no damage to the insert threads or heads, the insert shall not have rotated from its original position, and there shall be no permanent distortion of the shelter facility mounting surface or insert head or threads. The insert head (if any) shall be properly seated and tight against its bearing surface. Insert threads and outer face of the insert head, shall be free of paint.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:17Z
Check the source to verify that this is the currenversion before use.

3.11 Cleaning.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after final assembly of the units.

3.12 Finish.- The shelter facility shall be finished in accordance with Specification MIL-F-14072 and the drawings. (See 4.4).

3.13 Marking.-

3.13.1 General.- Marking shall conform to Specification MIL-M-13231 and the drawings. (See 4.4).

3.13.2 Nameplate.- Each shelter facility shall be provided with a nameplate in accordance with the drawings. Each nameplate shall be imprinted with a sequential serial number.

3.13.3 Aircraft loading data plate.- The shelter facility shall be provided with data plates in accordance with the drawings, imprinted with the dimensions which locate the centers-of-gravity of the shelter facility and the assemblage in the three reference planes; the gross weight of the shelter facility and assemblage; and the bearing pressure on the bottom of the skids. The shelter assemblage is the shelter facility with operational equipments or dummy loads in place. The locations of the centers-of-gravity shall be determined by the contractor using the preproduction sample.

3.13.4 Interior and exterior marking.- All interior and exterior marking of the shelter facility shall be as depicted on the drawings.

3.14 Riveting.- Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the currentversion before use.

3.15 Seal,vapor.- All joints and raw edges of the framing for the power, signal, video, and antenna entrance boxes, air-conditioner, and exhaust blowers shall be sealed to prevent entrance of moisture to the core material of the shelter panels and to the interior of the shelter. All joints shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws, and fasteners shall be dipped in sealer prior to installation. The sealer shall be one part Silicone Construction Sealant SE1204 as made by General Electric Company or equal.

3.16 Securing of parts.- Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain components of the facility securely so that loosening, wear of mounting arrangements, or permanent separation of parts or components will not occur when the equipment is subjected to the specified drop and transport tests.

3.16.1 Screws, threaded devices, and related parts.- Screws, threaded devices and related parts shall be as defined by the drawings. Thread forming screws shall not be used.

3.17 Soldering.-

3.17.1 Solder and flux.- Solder and flux shall be as defined on the drawings.

3.17.2 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.17.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, over heated or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:17Z
Check the source to verify that this is the currentversion before use.

3.18 Tropicalization of material.- Canvas,webbing,wood,etc shall be treated to be resistant to moisture and fungi as indicated on the drawings and as follows:

3.18.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.18.2 Toxicity.-Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.18.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.18.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2e).

3.19 Welding.- Welded joints shall be as defined by the drawings and shall be such that grinding on the finish weld will be unnecessary except where grinding is specified on the drawings. Inert-gas-shielded arc welding (MIG) using helium, argon, or a mixture of the two shall be used for arc welding of aluminum. Welds shall conform to MIL-W-6858 Class B and MIL-W-45205 Class B as applicable; surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free of scale, paint, grease, and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 4.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

3.20 Wiring and cabling.- Wiring and cabling shall be in accordance with the drawings and shall be neat and sturdy.

3.20.1 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided:

a. To prevent undue stress on cable forms, wires, and connections.

b. To facilitate field repair of broken or cut wires.

c. To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.20.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.20.3 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation unless specifically indicated by the drawings.

3.20.4 Crimping of solderless terminal lugs.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of Specification MIL-T-7928. (See 4.15).

3.20.5 Identification of wiring.- Wiring shall be identified by color-coding and wire pair marking, as indicated on the drawings.

3.21 Interchangeability.- Like units, assemblies,subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See 4.16). Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the current version before use.

3.22 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs: (See 4.17).

3.2	Construction
3.3	Electrical installation
3.10	Mounting inserts
3.11	Cleaning
3.12	Finish
3.13	Marking
3.14	Riveting
3.15	Seal,vapor
3.16	Securing of parts
3.17	Soldering
3.18	Tropicalization
3.19	Welding
3.20	Wiring and cabling
3.21	Interchangeability

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure that supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspection.- Inspection shall be classified as follows:

a. Preproduction inspection (does not include preparation for delivery). (See 4.3).

b. Inspection covered by subsidiary documents.(See 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5).

(2) Quality conformance inspection of preparation for delivery. (See 4.18).

4.3 Preproduction inspection.- This inspection shall be performed by the contractor on the preproduction unit unless otherwise specified in the contract. (See 3.1). It shall consist of the preproduction inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B, and Group C (except EVR) (see Inspection Tables I,II and III respectively). Preproduction inspection shall be performed in the following order, unless otherwise specified.

a. Mounting inserts
b. Electrical installation
c. System operation
d. Truck Transportability
e. Railroad transportability
f. Watertightness
g. Fording

The last tests performed shall be a repeat of the Electrical installation tests and System tests. Remaining preproduction tests specified for Groups A,B and C inspections but not included in the above list shall be performed in any order satisfactory to the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of preproduction inspection and inspection of equipment before preparation for delivery:

Item	Where required
Cable assemblies	3.3.6
Suppression,EMI	3.3.7
Finish	3.12
Marking	3.13
Welding	3.19

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the government - at its discretion - may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements (See 6.6). Test equipment for government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.- Each unit of equipment shall be inspected for conformance to all the inspection and test requirements of Table I of this specification. Equipment conforming to the above requirements shall be verified by the contractor,utilizing the procedures for the multilevel continuous sampling plan (CSP-M) of MIL-STD-1235, unless otherwise specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

Table I - Group A Inspection

Inspection	Rqr Para	Insp Para	AOQL % Def Major	AOQL % Def Minor
Visual and Mechanical	3.22	4.17	1.5 (AQL 1.0%)	5.0 (AQL 4.0%)
Electrical				
Power wiring				
Continuity	3.3.1.1	4.7.1.1		
Dielectric strength	3.3.1.2	4.7.1.2		
Insulation resistance	3.3.1.3	4.7.1.3		
Overvoltage control				
Dielectric strength	3.3.2.1	4.7.2.1		
Insulation resistance	3.3.2.2	4.7.2.2		
Voltage drop	3.3.2.3	4.7.2.3		
Power distribution	3.3.3	4.7.3 thru 4.7.3.3	1.5 (AQL 1.0%)** for the entire electrical wiring	
Signal wiring				
Continuity & circuit resistance	3.3.4.1	4.7.4.1		
Dielectric strength	3.3.4.2	4.7.4.2		
Insulation resistance	3.3.4.3	4.7.4.3		
Video wiring				
Continuity & circuit resistance	3.3.5.1	4.7.5.1		
Dielectric strength and insulation resistance	3.3.5.2	4.7.5.2		
Voltage standing wave ratio	3.3.5.3	4.7.5.3		

** All defects are major

NOTE: The order of inspection of Group A with respect to Groups B and C shall conform to the order specified in 4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

4.5.1.1 Group A verification sampling plan.- The Group A Verification Sampling plans for the AOQL's listed in Table I of this specification shall be selected from Tables VIII,IX and X of MIL-STD-1235 which are for f=½. The number of units in the Production Interval shall be the entire production quantity unless otherwise specified in the contract.

4.5.2 Group B inspection.- This inspection,including sampling, shall conform to Table II and to the procedures for multi-level continuous sampling plan (CSP-M) of Standard MIL-STD-1235,unless otherwise specified. Group B inspection shall be performed on units of production that have passed Group A inspection,except for Mounting Insert and Interchangeability Inspections which are performed prior. The government reserves the right to require complete retest of any sample which is reworked after failing Group B inspection.

Table II - Group B Inspection

Inspection	Rqr Para	Insp Para	AOQL % Def (see 4.5.2.1)
Mounting inserts	3.10	4.14.2	5.0(AQL 4.0%)
Interchangeability	3.21	4.16	5.0(AQL 4.0%)
Watertightness	3.8	4.12	3.0(AQL 2.5%)
Fording	3.9	4.13	3.0(AQL 2.5%)

NOTE: The order of inspection of Group B with respect to Groups A and C shall conform to the order specified in 4.3. The order of inspection within Group B shall be as in Table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

4.5.2.1 Group B sampling plans.- The Group B sampling plans for the AOQL's listed in Table II of this specification shall be selected from Tables VIII, IX and X of MIL-STD-1235 which are for f=½. The number of units in the Production Interval shall be the entire production quantity unless otherwise specified in the contract.

4.5.3 Group C inspection.- This inspection shall be as listed in Table III.

Table III - Group C Inspection

Inspection	Rqr Para	Insp Para
Solderless-terminal-lug connection	3.20.4	4.15
Construction verification	3.2	4.6
Mounting inserts	3.10	4.14.1
System operation	3.4	4.8
Truck Transportability	3.5	4.9
Railroad transportability	3.6	4.10

NOTE: The order of inspection of Group C with respect to Groups A and B shall conform to the order specified in 4.3. The order of inspection within Group C for all inspections listed shall be as listed in Table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

4.5.3.1 Sampling for inspection of construction evaluation.- The Equipment Verification Review, in accordance with paragraph 4.6 shall be performed once on each part and assembly, randomly selected, and on their installation into one of first 10 shelter facilities produced. This inspection shall not be performed on the preproduction unit.

4.5.3.2 Sampling for inspection of solderless-terminal-lug connections.- Five specimens of each combination of wire and terminal produced each week shall be prepared for inspection. A specimen is a 1 foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.3 Sampling for other inspection.- For inspection tests listed below, one unit shall be selected at random from the first 10 units or fraction thereof which have been produced, unless otherwise specified in the contract. Selection of the unit will be made by the government.

a. Mounting inserts
b. System operation
c. Truck transportability
d. Railroad transportability

4.5.3.4 Noncompliance.- The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and, (ii) all units or product which were manufactured under the same conditions and which the government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the QUALITY ASSURANCE REPRESENTATIVE. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed; at the option of the government) and Groups A and B inspection may be reinstituted; final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

4.5.4 Reinspection of conforming Group B and Group C sample units.- Unless otherwise specified, sample units which have been subjected to and pass Group B or Group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage (See 6.4).

4.6 Construction verification.- The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on order against the drawings (see 3.2) cited in the Technical Data Package (as supplemented by all approved engineering changes). EVR shall consist of the following:

a. An audit to establish that "as-built" shelter facilities, including individual parts, their use in subassemblies, and assemblies (components), and their final installation into the shelter forming the shelter facility, are in accordance with the end product drawings.

b. An audit to establish that "as-built" shelter facilities, including parts, components and their installation, meet the acceptance requirements specified in the Technical Data Package for each shelter part, components, and shelter facility. The audit shall record all acceptance test methods used together with resulting verification test data.

c. Preparation of EVR report(s) containing the following information:

(1) Identification of parts, components, shelter facilities and details proving compliance with 4.6a and b above.

(2) Discrepancies noted.

(3) Corrective action taken.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:17Z
Check the source to verify that this is the currentversion before use.

d. Two (2) copies of the EVR report shall be submitted no later than 15 days prior to the submission of the first production unit or lot to the government technical organization cited in the contract. The results of the EVR shall be subject to verification by authorized government personnel at the time the first unit or production lot is offered for acceptance. Government verification will be to the depth and extent necessary to demonstrate that the "as-built" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to government personnel during the verification. The contractor shall provide the following:

(1) Segregation of parts, components of shelter facilities, to permit reviewing government personnel access for detailed inspection. The contractor shall also provide personnel to disassemble any parts of the shelter facility necessary to permit government review of the shelter facility, its components and their installation, and to reassemble parts after inspection.

(2) Have as required, responsible personnel from each functional department available for discussions in their respective areas.

(3) Adequate administrative support for the EVR.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the current version before use.

4.7 Electrical installation tests.- The test of 4.7.1 through 4.7.5.3 shall be performed on each shelter facility prior to installation of other major components comprising the AN/TRC-151. Before applying power to the shelter facility, an electrical ground at the power entrance box shall be made. Power to the shelter facility shall be 115 volts AC, 60 cycles, single phase.

4.7.1 Power system and components installation tests.- Prior to conducting power wiring and components tests of 4.7.1.1, 4.7.1.2, and 4.7.1.3, the following shall be accomplished: (Refer to drawing SC-D-547753).

a. Remove or disconnect the following electrical loads:

(1) All fluorescent, incandescent, and neon (power indicator and power distribution box) lamps.

(2) All fluorescent lamp starters.

(3) Blowers and heaters.

b. Remove power distribution box cover panel and disconnect both conductors from the voltmeter terminals. Position or isolate disconnected leads so that they do not short to each other or to shelter facility ground.

c. Fabricate test cable assembly shown in Figure 1. Connect test cable between power cable connector marked "Input to CN-514" and cable connector marked "To Overvoltage Control".

4.7.1.1 Continuity of conductors.- Continuity resistance which meets the requirements of 3.3.1.1 shall be performed on each power circuit in accordance with the following paragraphs. A calibrated low resistance reading ohmmeter or equivalent shall be used in performing the test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the currentversion before use.

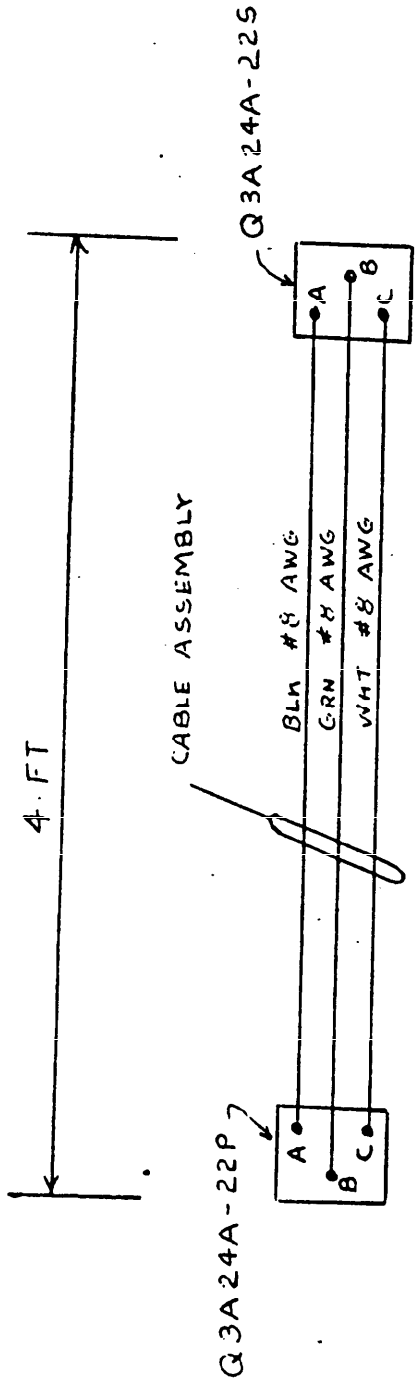


FIG. 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

a. Place the following circuits in the ON position:

(1) Fluorescent lights switch
(2) Incandescent cold start lights switch
(3) Blower-1 switch
(4) Blower-2 switch
(5) Main circuit breaker CB7
(6) Branch circuit breakers CB1 thru CB6
(7) System circuit breakers CB8 thru CB11

b. Place BLACKOUT/BYPASS switch to the BYPASS position.

c. Measure continuity resistance from pin 1 of power connector IN located at power entrance box to the following: brass (hot) contacts of outside power receptacle J1, Convenience power receptacles J2 and J3, Intercom power receptacle J4, Blowers power receptacles J5 and J6, Heaters power receptacles J7 and J8, and Equipment Systems power receptacles J9 thru J16; black wire termination at each fluorescent lamp; black wire termination at each incandescent lamp; and pin 1 of power OUT connector at power entrance box.

d. Place all controls listed under a above in the OFF position except main circuit breaker.

e. Measure continuity resistance from pin 2 of power connector IN to the following: silver (neutral) contacts of outside power receptacle J1, convenience power receptacles J2 and J3, Intercom power receptacle J4, Blowers power receptacles J5 and J6, Heaters power receptacles J7 and J8, and Equipment Systems power receptacles J9 thru J16; white wire termination at each fluorescent lamp; white wire termination at each incandescent lamp; and pin 2 of power OUT connector.

f. Place main circuit breaker in the OFF position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

g. Measure continuity resistance from pin 4 of power connector IN to the following: green (ground) contacts of outside power receptacle J1, Convenience power receptacles J2 and J3, Intercom power receptacle J4, Blowers power receptacles J5 and J6, Heaters power receptacles J7 and J8, and Equipment Systems power receptacles J9 thru J16; all ground terminals located at the power entrance box, power distribution box, video entrance box, and raceway; and pin 4 of power OUT connector.

h. Resistance measurements made in steps c,e, and g above, shall meet the requirements of 3.3.1.1.

4.7.1.2 Dielectric breakdown voltage.- Dielectric strength tests which meet the requirements of 3.3.1.2 shall be performed on each power circuit in accordance with the following paragraphs. A non-destructive DC dielectric tester shall be used in performing the test.

NOTE: Take preventive measures that will preclude personnel from coming into contact with conductors under test during performance of ALL dielectric and insulation resistance test. Discharge each conductor to ground following each test, to eliminate danger of shock.

a. Place all circuit controls listed under 4.7.1.1a in the ON position except FLUORESCENT lights switch.

b. Place BLACKOUT/BYPASS switch to the BYPASS position, and lock shelter facility door securely.

c. Test between pins 1 and 4 of power connector IN. Adjust tester for a value of not less than 500 volts DC output and maintain this potential for 30 seconds. The requirements of 3.3.1.2 shall be met.

d. Place FLUORESCENT LIGHTS switch to the ON position, and lock shelter facility door securely.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the currentversion before use.

e. Repeat step c above with tester connected first across pins 1 and 2, and then across pins 2 and 4.

4.7.1.3 Insulation resistance.- Insulation resistance which meets the requirement of 3.3.1.3 shall be made on each power circuit in accordance with the following paragraphs. A megohmmeter or equivalent shall be used in performing the test.

a. Repeat steps 4.7.1.2a thru e above except step 4.7.1.2c which shall be replaced by step b below.

b. Test between pins 1 and 4 of power connector IN. The output test voltage shall not be less than 100 volts DC nor more than 500 volts DC. Maintain this potential until the indicating meter has ceased fluctuating and indicates a steady or increasing insulation resistance. The requirements of 3.3.1.3 shall be met. (See Note under 4.7.1.2).

4.7.2 Overvoltage control tests.- Verification of the electrical requirements of 3.3.2 shall be made on each over-voltage power circuit as indicated by the following paragraphs: (Refer to drawing SC-C-532288).

4.7.2.1 Dielectric strength of insulation.-

a. Using a non-destructive DC dielectric tester, test between pins A and C of connector J1. Adjust tester for a value of not less than 500 volts DC output and maintain this potential for a period of 30 seconds. The requirements of 3.3.2.1 shall be met.

b. Repeat step a above with tester connected first across pins A and B and then across pins B and C of connector J1.

c. Repeat steps a and b above substituting connector J2 for connector J1.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the currentversion before use.

4.7.2.2 Resistance of insulation.-

a. Using a megohmmeter or equivalent, test between pins A and C of connector J1. Adjust for a value of not less than 100 volts DC nor more than 500 volts DC and maintain this potential until the indicating meter has ceased fluctuating and indicates a steady or increasing insulation resistance reading. The requirements of 3.3.2.2 shall be met.

b. Repeat step a above with tester connected first across pins A and B and then across pins B and C of connector J1.

c. Repeat steps a and b above substituting connector J2 for connector J1.

4.7.2.3 Voltage drop.-

a. Connect a resistive load to produce a current of approximately 50 amps ± 5 at connector J2, pins A and C.

b. Connect a 115 volt AC (nominal), 60 cycle supply across pins A and C of connector J1.

c. Turn unit switch to ON position. Neon lamp on unit shall light.

d. Measure voltage drop between pin A of J1 and Pin A of J2. The requirements of 3.3.2.3 shall be met.

e. Repeat step d above substituting pin C for pin A.

4.7.3 Power distributional and operational function tests.- Power distributional and operational function tests which meet the requirements of 3.3.3 shall be performed on each power circuit in accordance with the following paragraphs: (Refer to Drawing SC-D-547753).

4.7.3.1 Test preparation.-

a. Replace or reconnect all electrical loads and line voltmeter removed or disconnected by 4.7.1 a and b.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

b. Replace power distribution box cover.

c. Remove test cable assembly installed by 4.7.1c. Connect cable assembly marked "Input to CN-514" to input connector J1 of overvoltage control; and cable assembly marked "To Overvoltage Control" to output connector J2 of overvoltage control.

d. Place all circuit breakers and switches in the OFF position.

e. Place BLACKOUT/BYPASS switch to the BYPASS position.

f. Open outer exhaust port covers of both blowers.

g. Insert AC line plugs of both blowers into their respective receptacles.

h. Place a 115 volt AC test lamp with grounding plug into all convenience, equipment, intercom, and heater receptacles.

4.7.3.2 Test of line voltmeter and indicator lamp.-

a. Apply 115 volts AC (nominal), 60 cycles, single phase power to shelter facility at power connector IN. Power indicator neon lamp shall light and facility voltmeter at power distribution panel shall indicate power to the shelter.

b. Measure specific line voltage by connecting a calibrated AC voltmeter (0-150 volt scale) between pin 1 and pin 2 of power connector OUT.

c. Read voltage indicated on shelter facility voltmeter. The voltage read shall be within ±2% of the voltage measured by step b above.

d. Place main circuit breaker in the ON position.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:17Z
Check the source to verify that this is the currentversion before use.

4.7.3.3 Test of circuit breakers and switches.-

Circuit #1.- (Lights circuit breaker CB1)

a. Place lights circuit breaker CB1 in the ON position. Neon lamp above CB1 shall light.

b. Place FLUORESCENT LIGHTS switch in the ON position. All fluorescent lamps shall light.

c. Place INCANDESCENT COLD START LIGHTS switch in the ON position. All incandescent lamps shall light.

d. Place circuit breaker CB1 in the OFF position. All lamps mentioned in a thru c above shall go out.

e. Replace circuit breaker CB1 in the ON position to restore lights.

f. With shelter facility door open, place BLACKOUT/BYPASS switch to the BLACKOUT position. All fluorescent and incandescent lamps shall go out.

g. Close shelter facility door. All fluorescent and incandescent lamps shall light.

h. Place FLUORESCENT LIGHTS switch in the OFF position. All fluorescent lamps shall go out.

i. Place INCANDESCENT COLD START LIGHTS switch in the OFF position. All incandescent lamps shall go out.

j. Open door and restore lights by placing BLACKOUT/BYPASS switch to BYPASS and FLUORESCENT LIGHTS switch to the ON position.

Circuit #2.- (Blowers circuit breaker CB2)

a. Place Blowers circuit breaker CB2 in the ON position. Neon lamp above CB2 shall light.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

b. Place roadside blower switch in the ON position. Roadside blower shall operate and shall exhaust air from shelter. The spring loaded exhaust port cover shall open by the force of the exhaust blower.

c. Repeat step b above for curbside switch and blower.

d. Place circuit breaker CB2 in the OFF position. Neon lamp above CB2 shall go out; both blowers shall cease to operate; and the spring loaded port covers shall close.

Circuit #3.- (Heater-1 circuit breaker CB3)

a. Place Heater-1 circuit breaker CB3 in the ON position. Neon lamp above CB3 shall light and test lamp in heater-1 receptacle shall light.

b. Place circuit breaker CB3 in the OFF position. All lamps mentioned in step a above shall go out.

Circuit #4.- (Convenience and Intercom circuit breaker CB4)

a. Place convenience and intercom circuit breaker CB4 in the ON position. Neon lamp above CB4 and test lamps in each convenience and intercom receptacle shall light.

b. Place circuit breaker CB4 in the OFF position. All lamps mentioned in step a above shall go out.

Circuit #5.- (Heater-2 circuit breaker CB6)

a. Place heater-2 circuit breaker CB6 in the ON position. Neon lamp above CB6 and test lamp in heater-2 receptacle shall light.

b. Place circuit breaker CB6 in the OFF position. All lamps mentioned in step a above shall go out.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the currentversion before use.

Circuit #6.- (Voltage Regulator circuit breaker CB6)

a. Place voltage regulator circuit breaker in the ON position. Neon lamp above CB6 shall light. Neon lamp on overvoltage control shall not light.

b. Place switch of overvoltage control in the ON position. Neon lamp on this unit shall light.

c. Place circuit breaker CB6 in the OFF position. Neon lamp on overvoltage control shall go out.

Circuit #7.- (Main circuit breaker CB7)

a. Place all circuit breakers, except main circuit breaker in the ON position. Place all switches in the ON position. Place BLACKOUT/BYPASS switch in the BYPASS position.

b. Place main circuit breaker CB7 in the ON position. All neon lamps above circuit breakers shall light. All fluorescent and incandescent lamps shall light. Test lamps in all convenience intercom, and heater receptacles shall light. Both blowers shall operate. Test lamps in all equipment receptacles shall not light.

c. Place switch on overvoltage control unit in the ON position. Neon lamp on this unit shall light and test lamps in all equipment receptacles shall light.

d. Place main circuit breaker in the OFF position. All lamps and blowers mentioned in b and c above shall go out.

e. Place main circuit breaker in the ON position. Condition of step b above shall exist only.

f. Place all circuit breakers in the OFF position except the main, lights, and voltage regulator circuit breakers. Place overvoltage control switch to the ON position.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the currentversion before use.

Circuit #8.- (Radio system 1 circuit breaker CB8.)

a. Place TO SYS-1 circuit breaker CB8 in the ON position. Test lamp in radio system 1 receptacle shall light.

b. Place circuit breaker CB8 in the OFF position. Test lamp mentioned in step a above shall go out.

Circuit #9.-(Radio system 2 circuit breaker CB9.)

a. Place TO SYS-2 circuit breaker CB9 in the ON position. Test lamp in radio system 2 receptacle shall light.

b. Place circuit breaker CB9 in the OFF position. Test lamp mentioned in step a above shall go out.

Circuit #10.-(MUX system 1 circuit breaker CB10.)

a. Place TO SYS-1 circuit breaker CB10 in the ON position. Test lamps in system 1 MUX receptacles shall light.

b. Place circuit breaker CB10 in the OFF position. Test lamps mentioned in step a above shall go out.

Circuit #11.- (MUX system 2 circuit breaker CB11.)

a. Place TO SYS-2 circuit breaker CB11 in the ON position. Test lamps in system 2 MUX receptacles shall light.

b. Place circuit breaker CB11 in the OFF position. Test lamps mentioned in step a above shall go out.

4.7.4 Signal (audio) system wiring and components installation tests.- (Refer to Drawings SC-D-539634 and SC-D-547769).

4.7.4.1 Continuity of conductors.- Continuity resistance which meets the requirements of 3.3.4.1 shall be performed on each audio-line circuit in accordance with the following paragraphs. A low resistance continuity tester or equivalent shall be used. (An automatic tester may be used, subject to government approval).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

a. Place all CABLE/BINDING POST channel switches in the BINDING POST position.

b. Connect tester between conductor A of audio connector marked "SYSTEM 1 CHANNELS 1-4" and contact 2B of signal connector, system 1, located at signal entrance box. No circuit continuity shall exist.

c. Place channel switch S1 of system 1 to CABLE position. Circuit continuity shall now exist and meet the requirements of 3.3.4.1. Replace channel switch S1 to BINDING POST position.

d. Repeat steps b and c above for each of the remaining conductors of the same audio connector to the corresponding contacts of the same signal connector; each time placing to CABLE position the channel switch correlating with the circuit being tested and replacing to BINDING POST position following test. (Refer to drawings in 4.7.4 for circuit path from audio to signal (connectors).

e. Repeat steps b thru d above for audio connectors marked "SYSTEM 1 CHANNELS 5-8" and "SYSTEM 1 CHANNELS 9-12" to the same signal connector as mentioned in step b above.

f. Repeat steps b thru e above for all remaining audio connectors of system 2 corresponding to system 2 signal connector.

g. Place all CABLE/BINDING POST channel switches in the CABLE position.

h. Connect tester between conductor A of audio connector marked "SYSTEM 1 CHANNELS 1-4" and binding post 2B of system 1 located at signal entrance box. No circuit continuity shall exist.

i. Place channel switch S1 of system 1 to BINDING POST position.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the currentversion before use.

j. Repeat steps h and i above for each of the remaining conductors of the same audio connectors to be the corresponding binding posts of system 1; each time placing to BINDING POST position the channel switch correlating with the circuit being tested and replacing to CABLE position following test. (Referenced drawings in 4.7.4).

k. Repeat steps h thru j above for the two remaining audio connectors of system 1 to the corresponding binding posts of system 1.

l. Repeat steps h thru k above for all remaining audio connectors of system 2 corresponding to binding posts of system 2.

m. Connect phone/intercom parallel interconnecting cable assembly between the phone/intercom plugs located on power entrance panel and raceway.

n. Connect tester across the phone binding posts at power entrance box. No continuity shall exist. Repeat for intercom binding posts, order wire system 1 binding posts and order wire system 2 binding posts.

o. Insert a shorted plug at phone jack. Connect tester across the phone binding posts at power entrance box. Circuit continuity shall exist and meet the requirements of 3.3.4.1.

p. Repeat step o above for the intercom jack and binding posts.

q. Connect tester between following points. Circuit continuity shall exist and meet the requirements of 3.3.4.1:

(1) From phone A binding post located at power entrance box to:

(a) Binding posts 25A of systems 1 and 2.

(b) Contacts 25A of systems 1 and 2 signal connector.

(2) From phone B binding post located at power entrance box to:

(a) Binding posts 25B of systems 1 and 2.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the currentversion before use.

(b) Contacts 25B of systems 1 and 2 signal connectors.

(3) From intercom A binding post located at power entrance box to:

(a) Binding Posts 26A of systems 1 and 2.

(b) Contacts 26A of systems 1 and 2 signal connectors.

(4) From intercom B binding post located at power entrance box to:

(a) Binding Posts 26B of systems 1 and 2.

(b) Contacts 26B of systems 1 and 2 signal connectors.

r. Disconnect phone/intercom interconnecting cable assembly and repeat steps q(1) thru q(4) above. No continuity shall exist.

s. Connect tester between system-1 remote phone A binding post at power entrance box and one of the unterminated system-1 remote phone wires to indicate circuit continuity.

t. Connect tester between system-1 remote phone B binding post at power entrance box and the other unterminated system-1 remote phone wire to indicate continuity.

u. Repeat steps s and t above for system-2 remote phone lines.

v. Using the RADIO TO CABLE PATCH THRU cable assembly connect between AN/GRC-50 PATCH THRU SYS-1 and TD-204 or TD-754 PATCH THRU SYS-1 at patch panel.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the currentversion before use.

w. Measure continuity resistance between the following pins of cable assemblies marked RAD HANDSET O.W. SYS-1 and TD-204/754 PATCH THRU SYS-1: C to E, E to F, A to G, and B to H respectively.

x. Repeat steps v and w above for system 2.

y. Using the CABLE REPEATER PATCH THRU cable assembly connect between TD-204 or TD-754 PATCH THRU SYS-1 and TD-204 or TD-754 PATCH THRU SYS-2 at patch panel.

z. Measure continuity resistance between the following pins of cable assemblies marked TD-204/754 PATCH THRU SYS-1 and TD-204/754 PATCH THRU SYS-2: E to A, F to B, A to E, B to F respectively; pins C to D of TD-204/754 PATCH THRU SYS-1; and pins C to D of TD-204/754 PATCH THRU SYS-2.

aa. Using the RADIO TO CABLE ORDER WIRE patch cable connect between AN/GRC-50 ORDER WIRE SYS-1 and TD-202 ORDER WIRE SYS-1 at patch panel.

ab. Measure continuity resistance between the following pins of cable assemblies marked RAD ORDER WIRE SYS-1 and TD-202 ORDERWIRE SYS-1: A to A, B to B, C to C, and D to D.

ac. Repeat step ab above for system 2.

ad. Circuit continuity measurements made in steps s,t,u,w,x,z,ab, and ac above shall meet the requirements of 3.3.4.1.

4.7.4.2 Dielectric breakdown voltage.- Dielectric strength tests which meet the requirements of 3.3.4.2 shall be performed on each audio-line circuit in accordance with the following paragraphs. A non-destructive DC dielectric test instrument and suitable test jigs shall be used in performing the test (an automatic tester may be used, subject to government approval).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

a. Using appropriate test jigs or test fixtures, connect the following to shelter facility ground terminal:

(1) All contacts, 1A thru 26B, of signal connectors, systems 1 and 2.

(2) All binding posts in the signal and power entrance boxes.

(3) All shields of the six audio connectors of systems 1 and 2.

(4) All pins of each connector at the PATCH THRU and ORDER WIRE PATCH PANEL.

b. Place all CABLE/BINDING POST channel switches in the CABLE position.

c. Connect tester between shelter facility ground terminal and conductor A of audio connector marked "SYSTEM 1 CHANNELS 1-4". Remove ground connection from contact 2B of signal connector, system 1.

d. Adjust tester to 500 volts DC output and maintain this potential for a continuous period of not less than 5 seconds. The requirements of 3.3.4.2 shall be met. Re-establish ground connection(s) to the contact(s) or binding post(s) under test.

e. Repeat steps c and d above for each of the remaining conductors of the same audio connector; each time ungrounding the contact of signal connector correlating with the circuit being tested and reconnecting to shelter facility ground following completion of test. (Referenced drawings in 4.7.4).

f. Repeat steps c thru e above for each of the remaining audio connectors of system 1 and 2 corresponding to signal connectors 1 and 2 respectively.

g. Place all CABLE/BINDING POST channel switches in the BINDING POST position.

h. Repeat steps c thru f substituting binding posts of systems 1 and 2 for contacts of audio connectors 1 and 2 respectively.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

i. Connect tester between shelter facility ground terminal and Phone A binding post at power entrance box. Remove ground connections from contacts 25A of signal connectors 1 and 2; binding posts 25A of systems 1 and 2; and phone A binding post at power entrance box.

j. Repeat step d above with a potential of 250 volts DC output.

k. Repeat steps i and j above for the other Phone,Intercom, and Remote Phone binding posts at power entrance box, and shelter facility ground; each time ungrounding all contacts and binding posts correlating with the circuit being tested and reconnecting to shelter facility ground terminal following completion of test.

l. Place tester between Pin C of cable assembly marked RAD HANDSET O.W. SYS-1 and shelter facility ground terminal. Remove ground connection from pin E of AN/GRC-50 PATCH THRU SYS-1 connector at patch panel.

m. Adjust tester to 500 volts DC output and maintain this potential for a continuous period of not less than 5 seconds. The requirements of 3.3.4.2 shall be met. Re-establish ground connection to the pin under test.

n. Repeat steps l and m above for the following pins of cable assembly RAD HANDSET O.W. SYS-1: E,A, and B; each time ungrounding the pin of connector AN/GRC-50 PATCH THRU SYS-1 correlating with the circuit being tested as follows: F,G, and H respectively. After each test reconnect the pin being tested to shelter facility ground terminal.

o. Repeat steps l thru n above substituting system 2 for system 1.

p. Place tester between pin A of cable assembly marked TD-204/754 PATCH THRU SYS-1 and shelter facility ground terminal. Remove ground connection from pin A of TD-204 or TD-754 PATCH THRU connector at patch panel.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:17Z
Check the source to verify that this is the current version before use.

q. Adjust tester to 500 volts DC output and maintain this potential for a continuous period of not less than 5 seconds. The requirements of 3.3.4.2 shall be met. Re-establish ground connection to the pin under test.

r. Repeat steps p and q above for the following pins of cable assembly TD-204/754 PATCH THRU SYS-1: B,C,D,E,F,G, and H; each time ungrounding the pin of connector TD-204 or TD-754 PATCH THRU SYS-1 correlating with the circuit being tested as follows: B,C,D,E,F,G, and H respectively. After each test reconnect the pin being tested to shelter facility ground terminal.

s. Repeat steps p thru r above substituting system 2 for system 1.

t. Place tester between pin A of cable assembly marked RAD ORDER WIRE SYS-1 and shelter facility ground terminal. Remove ground connection from pin A of AN/GRC-50 ORDER WIRE connector at patch panel.

u. Adjust tester to 500 volts DC output and maintain this potential for a continuous period of not less than 5 seconds. The requirements of 3.3.4.2 shall be met. Re-establish ground connection to the pin under test.

v. Repeat steps t and u above for the following pins of cable assembly RAD ORDER WIRE SYS-1: B,C, and D: each time ungrounding the pin of connector AN/GRC-50 ORDER WIRE SYS-1 correlating with the circuit being tested as follows: B,C, and D respectively. After each test reconnect the pin being tested to shelter facility ground terminal.

w. Repeat steps t thru v above substituting system 2 for system 1.

x. Repeat steps t thru w above substituting TD-202 ORDER WIRE SYS-1 for RAD ORDER WIRE SYS-1 and TD-202 ORDER WIRE SYS-1 connector for AN/GRC-50 ORDER WIRE SYS-1 connector.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the currentversion before use.

4.7.4.3 Insulation resistance.- Insulation resistance which meets the requirements of 3.3.4.3 shall be performed on each audio-line circuit in accordance with the following paragraphs. A megohmmeter tester or equivalent shall be used in performing the test. (An automatic tester may be used, subject to government approval).

a. Repeat steps 4.7.4.2a thru x above, except 4.7.4.2d, j,m,q, and u which shall be replaced by step b below.

b. Adjust tester to a value of not less than 100 nor more than 500 volts DC output. Maintain this potential for one minute or until the indicating meter has ceased fluctuating and indicates steady or increasing insulation resistance. The requirements of 3.3.4.3 shall be met. Re-establish ground connection(s) to the contact(s), binding post(s) or pin(s) under test.

4.7.5 Video and radio frequency system wiring and components installation.- (Refer to Signal Schematic Drawing SC-F-585142).

4.7.5.1 Continuity of conductors.- Continuity resistance which meets the requirements of 3.3.5.1 shall be performed on each video and RF circuit in accordance with the following paragraphs. A calibrated low resistance reading ohmmeter or equivalent shall be used in performing the test.

a. Circuit continuity resistance which meets the requirements of 3.3.5.1 shall exist when tested in accordance with Table IV. The method of testing shall be inner conductor, to inner conductor and outer conductor (shield) to outer conductor (shield).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

Table IV - Continuity Measurements

FROM PATCH PANEL MARKED	TO CABLE ASSEMBLY MARKED
TD-660 PCM OUT SYS-1	TD-660 PCM OUT SYS-1
TD-660 PCM IN SYS-1	MUX PCM IN SYS-1
TD-660 TIM IN SYS-1	MUX TIM IN SYS-1
TD-660 TIM OUT SYS-1	MUX TIM OUT SYS-1
TD-660 ALT PCM OUT SYS-1	ALT PCM OUT SYS-1
TD-660 SYNC OUT XMTR SYS-1	SYNC OUT XMTR SYS-1
TD-660 SYNC IN SYS-1	SYNC IN SYS-1
TD-660 SYNC OUT RCVR SYS-1	SYNC OUT RCVR SYS-1
TD-202 PCM OUT SYS-1	TD-202 PCM OUT-1 SYS-1
TD-202 PCM IN SYS-1	TD-202 PCM IN-1 SYS-1
TD-202 TIM IN SYS-1	TD-202 TIM IN SYS-1
TD-202 TIM OUT SYS-1	TD-202 TIM OUT-1 SYS-1
TD-202 PCM IN-2 SYS-1	TD-202 PCM IN-2 SYS-1
TD-202 PCM OUT-2 SYS-1	TD-202 PCM OUT-2 SYS-1
TD-202 TIM OUT-2 SYS-1	TD-202 TIM OUT-2 SYS-1
TD-204 PCM OUT SYS-1	COMB PCM OUT-1 SYS-1
TD-204 PCM IN SYS-1	COMB PCM IN-1 SYS-1
TD-204 TIM IN SYS-1	COMB TIM IN SYS-1
TD-204 TIM OUT SYS-1	COMB TIM OUT-1 SYS-1
TD-204 PCM IN-2 SYS-1	COMB PCM IN-2 SYS-1
TD-204 PCM OUT-2 SYS-1	COMB PCM OUT-2 SYS-1
TD-204 TIM OUT-2 SYS-1	COMB TIM OUT-2 SYS-1
TD-204 FROM CABLE SYS-1	TD-754/TD-204 FROM CABLE SYS-1
TD-204 TO CABLE SYS-1	TD-754/TD-204 to CABLE SYS-1
VID ENT FROM CABLE SYS-1	VID ENT BOX MALE(M) SYS-1
VID ENT TO CABLE SYS-1	VID ENT BOX FEMALE(F) SYS-1
AN/GRC-50 RADIO IN SYS-1	RAD PCM IN SYS-1
AN/GRC-50 RADIO OUT SYS-1	RAD PCM OUT SYS-1
TD-202 RADIO IN SYS-1	TD-202 FM RAD RCVR SYS-1
TD-202 RADIO OUT SYS-1	TD-202 to RAD XMTR SYS-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

b. Repeat continuity check of step a above substituting system 2 for system 1.

c. Measure continuity resistance between the inner conductor of antenna connector, system 1, and inner conductor of cable assembly marked to XMTR of system 1. The requirements of 3.3.5.1 shall be met.

d. Repeat step c above for the shields of the same circuit.

e. Repeat steps c and d above substituting system 2 for system 1.

4.7.5.2 Dielectric and insulation resistance tests.- Dielectric and insulation resistance tests which meets the requirements of 3.3.5.2 shall be performed on each video and RF circuit in accordance with the following paragraphs. A non-destructive dielectric and insulation resistance tester shall be used in performing the test.

a. At Cable Patch Panel make the following cable connections using Cable Patch Cords:

Video ENT BOX SYS-1	TD-204 SYS-1
FROM CABLE	FROM CABLE
TO CABLE	TO CABLE

VIDEO ENT BOX SYS-2	TD-204 SYS-2
FROM CABLE	FROM CABLE
TO CABLE	TO CABLE

b. Remove all lightning protector elements at video entrance box.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

c. Apply test potential of 1500 volts DC for not less than 30 seconds between the following pin connections and terminal ground located at video entrance box. The requirements of 3.3.5.2 shall be met.

(1) System 1 male

(2) System 1 female

(3) System 2 male

(4) System 2 female

d. Replace lightning protector elements removed by step b above.

e. Repeat step c above applying test potential of 1100 volts DC.

f. Connect tester between inner conductor of antenna connector, system 1, and shelter facility ground terminal.

g. Adjust tester to 1700 RMS output and maintain this potential for a continuous period of not less than 30 seconds. The requirements of 3.3.5.2 shall be met.

h. Repeat steps f and g above substituting system 2 for system 1.

i. Apply test potential of 1500 volts DC for not less than 5 seconds between inner and outer (shield) conductor of each RF connector at the Video and Radio Patch Panel. The requirements of 3.3.5.2 shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

4.7.5.3 Voltage standing wave ratio (VSWR).- VSWR, which meets the requirements of 3.3.5.3 shall be made on each RF circuit in accordance with the following paragraphs. The VSWR measuring set shall consist of the following calibrated test equipment, or equal.

Hewlett-Packard Signal Generator HP-612A
Hewlett-Packard Signal Generator HP-614A
Hewlett-Packard Square Wave Generator HP-211A
Hewlett-Packard SWR Meter HP-415D
Hewlett-Packard Slotted Line HP-805C
Weinschel Model 535-FC 50 ohm coaxial termination, Female Type "C" connector
Cable Assembly Adapter,Female Type "C" to Male "QL" Connector
Ancillary items, including interconnecting cables, adapters, and any additional equipment needed.

a. Connect the test equipment using appropriate interconnecting cables and adapters as shown in Figure 2 to system 1 antenna circuit.

b. Set signal generator to a frequency of 600 megahertz.

c. Set the square wave generator for square wave output.

d. Set all test equipment for proper operation and obtain a VSWR reading on the SWR meter.

e. The VSWR reading obtained shall meet the requirements of 3.3.5.3.

f. Repeat steps b thru e above for signal generator frequencies of 1000 megahertz, 1350 megahertz, and 1850 megahertz respectively.

g. Repeat steps a thru f above for system 2 antenna circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the current version before use.

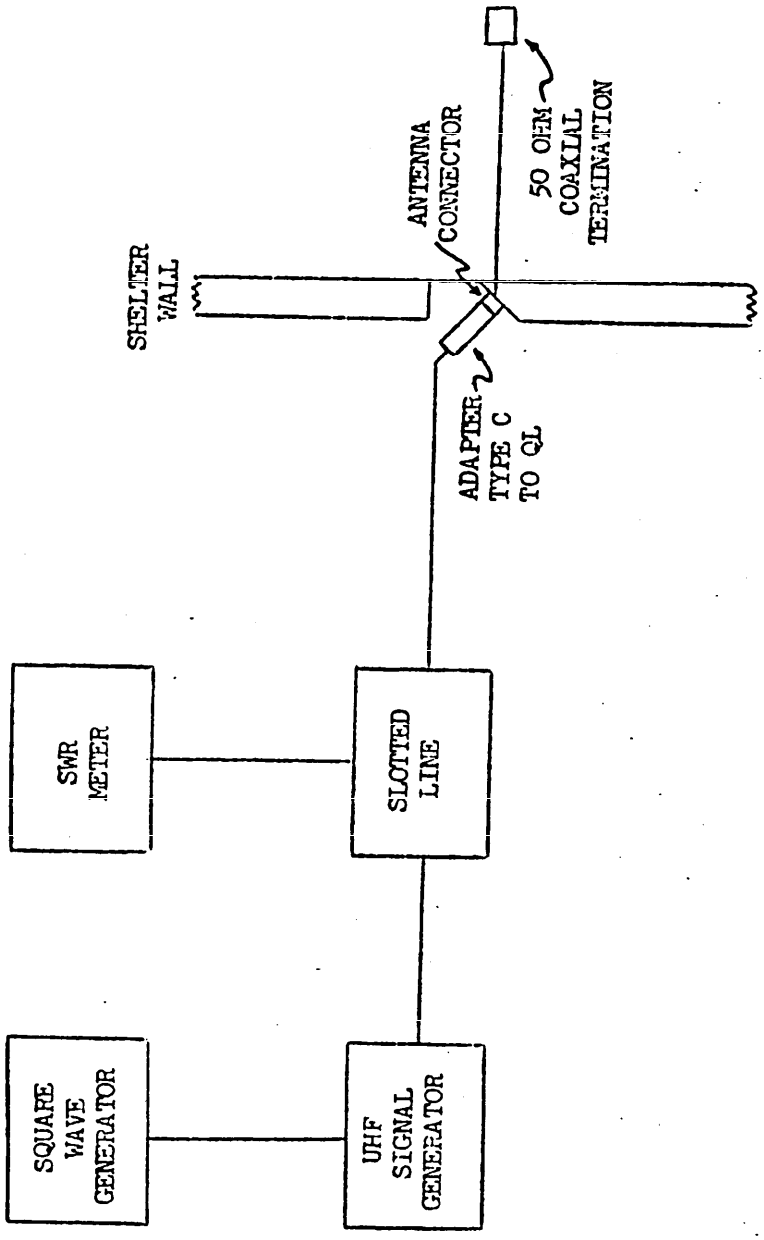


FIG. 2 VSWR TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

4.8 System test.- Install operational equipment in their respective rack locations and connect all power, signal, and radio frequency (RF) cables to designated receptacles and connectors. (See 3.4) (Refer to signal schematic plate).

4.8.1 Radio Terminal Test.- (See Figure 3)

a. At Video Patch Panel make the following cable connections using Video patch cords:

TD-660 SYS-1	TD-202 SYS-1
PCM IN	PCM OUT
PCM OUT	PCM IN
TIM IN	TIM OUT
TIM OUT	TIM IN

TD-660 SYS-2	TD-202 SYS-2
PCM IN	PCM OUT
PCM OUT	PCM IN
TIM IN	TIM OUT
TIM OUT	TIM IN

b. At Order Wire Patch Panel make the following cable connections using the Radio/Cable Order Wire patch cords: AN/GRC-50 Order Wire System 1 to TD-202 Order Wire System 1; and AN/GRC-50 Order Wire System 2 to TD-202 Order Wire System 2.

c. At Radio Patch Panel make the following cable connections using Radio Patch Cords:

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the currentversion before use.

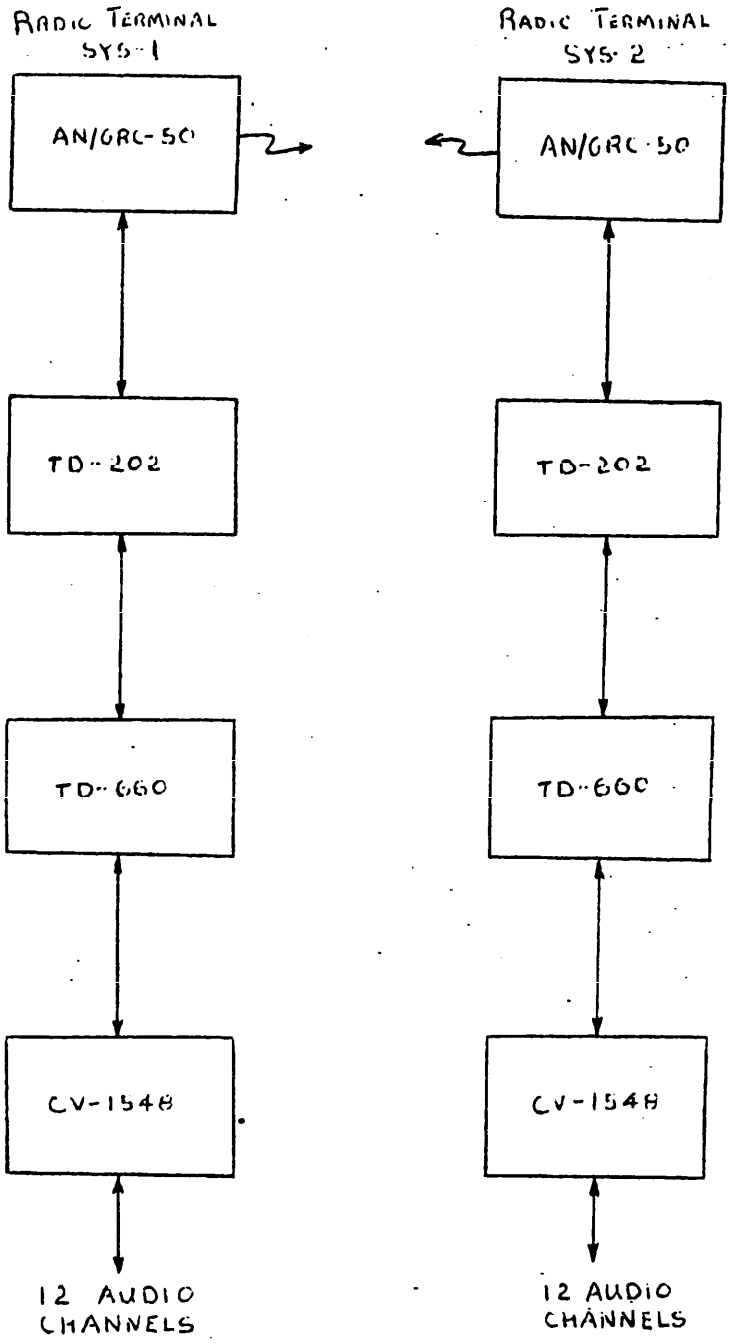


FIG. 3 TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

AN/GRC-50 SYS-1	TD-202 SYS-1
RADIO IN	RADIO OUT
RADIO OUT	RADIO IN
AN/GRC-50 SYS-2	**TD-202 SYS-2**
RADIO IN	RADIO OUT
RADIO OUT	RADIO IN

d. Connect Cable Assemblies CX-4566/U (25 ft) to each signal connector at signal entrance box. Connect other ends of the cable assemblies to two junction boxes J-1077A/U.

e. Place all channel switches at signal entrance panel to CABLE position.

f. Connect Telephone Sets TA-312/PT to pair 1 of each junction box. Phone 1 shall correspond to system 1 and Phone 2 to system 2.

g. Set channel switches of each CV-1548 to 2W; and signaling mode switches of each CV-1548 to AC.

h. Set 2 wire/4 wire switch of each TD-660 to 2 wire; MASTER/SLAVE switch to MASTER; and 6 CH/12 CH mode to 12 CH.

i. Set Traffic Sel switch of each TD-202 to 12; and OPR TEST switch to OPR.

j. Connect cable assemblies CG-1859/U (40 ft) to each antenna connector at antenna entrance box. Connect other ends to cable assemblies CG-718()/U (6 ft) using adapters UG-1374()/U. Connect other end of cable assemblies CG-718()/U to Dummy Loads DA-189/GRC. Separate the Dummy Loads to a distance of about 5 feet.

k. Set Converters CV-1548, Multiplexers TD-660, Combiners TD-202, and Radio Sets AN/GRC-50 for proper operations in accordance with the provided instructions. The dummy loads may be moved accordingly to receive a good signal strength.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

NOTE: The transmit and receive frequencies of the AN/GRC-50 must be separated by at least 15 channels.

1. Connect Handset H-156/U to each radio receiver. Ring radio system 1 from radio system 2; ring radio system 2 from radio system 1; establish communications.

m. At junction boxes ring phone 1 from phone 2; ring phone 2 from phone 1; establish communications.

n. Repeat step m above with phones connected to each odd binding post thru pair 23 on each J-1077A/U.

o. Set all channel switches on signal entrance panel to BINDING POST position.

p. Connect phone 1 and 2 to binding post pair 2WT/R CH-1 for systems 1 and 2 respectively.

q. Ring phone 1 from phone 2; ring phone 2 from phone 1; establish communications.

r. Repeat step q above with phones connected to binding posts 2WT/R for the remaining channels of systems 1 and 2.

s. Arrange switches in all CV-1548's and TD-660's for 4-wire operation.

t. Connect phone 1 to binding post pair 2 WT/R, CH-1, system 1; and phone 2 to binding post pair 4WT,CH-1, system 2, Receive communications at phone 1 from phone 2.

u. Connect phone 1 to binding post pair 4WT, CH-1, system 1; and phone 2 to binding post pair 2WT/R, CH-1, system 2, Receive communication at phone 2 from phone 1.

v. Repeat steps t and u above for all remaining channels of systems 1 and 2.

w. Place all channel switches at signal entrance panel to CABLE position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

x. At junction boxes, receive communications at phone 1 from phone 2 with each of the following phone connections.

PHONE 1 Binding Post Pair on J-1077A/U	PHONE 2 Binding Post Pair on J-1077A/U
1	2
3	4
5	6
7	8
9	10
11	12
13	14
15	16
17	18
19	20
21	22
23	24

y. Receive communications at phone 2 from phone 1 with each of the following phone connections:

PHONE 1 Binding Post Pair on J-1077A/U	PHONE 2 Binding Post Pair on J-1077A/U
2	1
4	3
6	5
8	7
10	9
12	11
14	13
16	15
18	17
20	19
22	21
24	23

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the current version before use.

4.8.2 Cable Terminal Test.- (See Figure 4)

a. Disconnect all patch cables at patch panel.

b. At Video Patch Panel make the following cable connections using Video patch cords:

TD-660 SYS-1	TD-204 SYS-1
PCM IN	PCM OUT
PCM OUT	PCM IN
TIM IN	TIM OUT
TIM OUT	TIM IN

TD-660 SYS-2	TD-204 SYS-2
PCM IN	PCM OUT
PCM OUT	PCM IN
TIM IN	TIM OUT
TIM OUT	TIM IN

c. At Cable Patch Panel make the following cable connections using Cable patch cords:

VIDEO ENT BOX SYS-1	TD-204 SYS-1
FROM CABLE	FROM CABLE
TO CABLE	TO CABLE

VIDEO ENT BOX SYS-2	TD-204 SYS-2
FROM CABLE	FROM CABLE
TO CABLE	TO CABLE

d. Interconnect system 1 and system 2 video connectors at video entrance box using cable assembly CX-11230/G.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

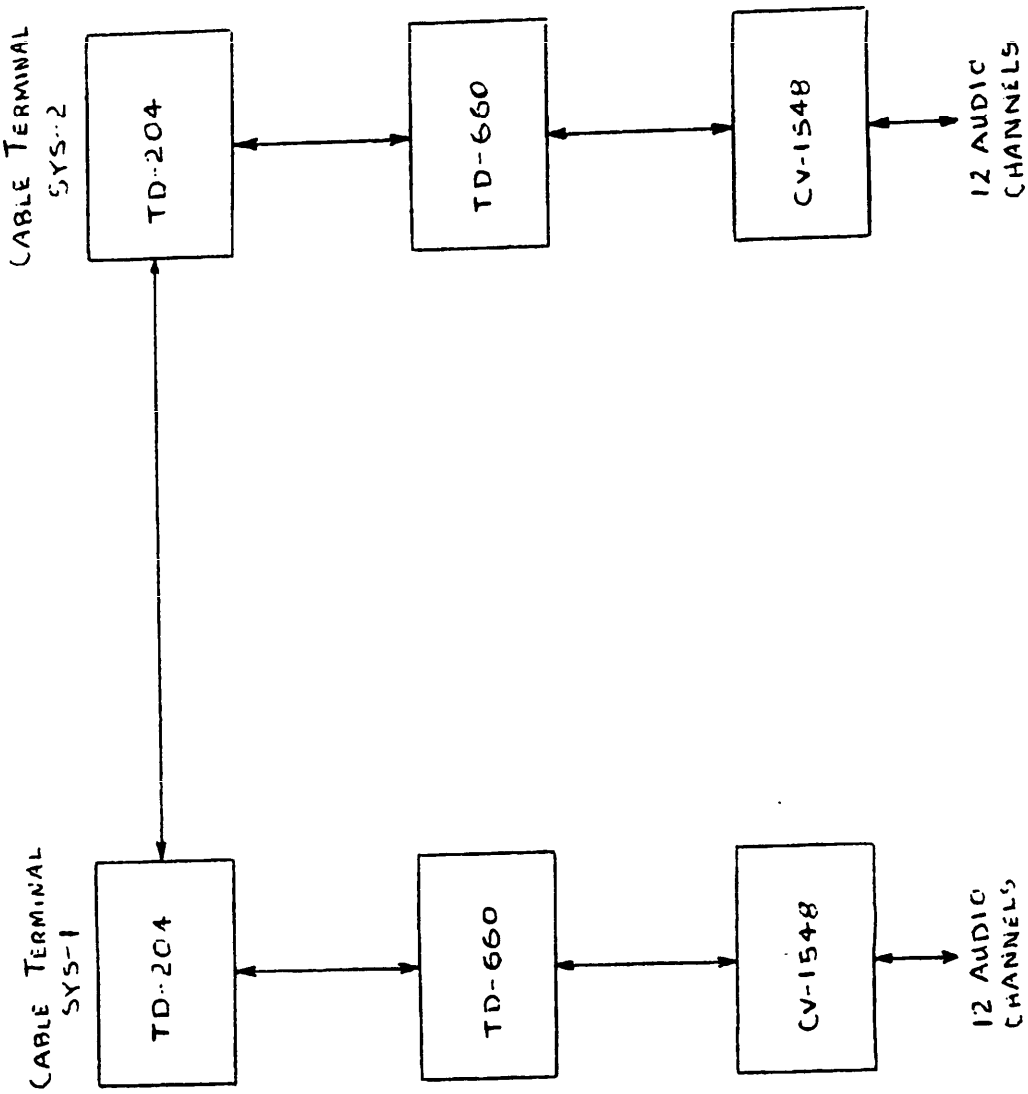


FIG. 4 TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

e. Set 2 wire/4 wire switch of each TD-660 to 2 wire; MASTER/SLAVE switch to MASTER; and 6 CH/12 CH mode to 12 CH.

f. Set each channel switch of each CV-1548 to 2 W.

g. Set Traffic Sel switch of each TD-204 to 12; READ/ ZERO SET/NORM OPR switch to NORM OPR;and the MILE switches on panels 6A4 and 6A5 to 1/2.

h. Set Converters CV-1548, Multiplexers TD-660, and Combiners TD-204 for proper operations in accordance with the provided manuals.

i. Connect Headsets H-91A/U to both combiners. Ring combiner 1 from combiner 2; ring combiner 2 from combiner 1; establish communications.

j. Place all channel switches at signal entrance panel to BINDING POST position.

k. Connect phone 1 and phone 2 to binding post pair 2WT/R CH-1 for systems 1 and 2 respectively.

l. Ring phone 1 from phone 2; ring phone 2 from phone 1; establish communications.

m. Repeat steps m and l above for channels 7 and 12.

n. Arrange switches in all CV-1548's TD-660's for 4-wire operation.

o. Connect phone 1 to binding post pair 2WT/R, CH-1, system 1; connect phone 2 to binding post pair 4WT, CH-1, system 2. Receive communications at phone 1 from phone 2.

p. Connect phone 1 to binding post pair 4WT, CH-1, system 1; and phone 2 to binding post pair 2WT/R, CH-1, system 2. Receive communication at phone 2 from phone 1.

q. Repeat steps o and p above for channels 6 and 11.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the currentversion before use.

4.8.3 RADIO/CABLE TERMINAL TEST.- (See Figure 5)

a. Disconnect all patch cables at patch panel.

b. At Video Patch Panel make the following cable connections using Video patch cords:

TD-660 SYS-1	TD-202 SYS-1
PCM IN	PCM OUT
PCM OUT	PCM IN
TIM IN	TIM OUT
TIM OUT	TIM IN
TD-204 SYS-1	**TD-660 SYS-2**
PCM IN	PCM OUT
PCM OUT	PCM IN
TIM IN	TIM OUT
TIM OUT	TIM IN
TD-202 SYS-2	**TD-204 SYS-2**
PCM IN	PCM OUT
PCM OUT	PCM IN
TIM IN	TIM OUT
TIM OUT	TIM IN

c. At Patch Thru Patch Panel make the following cable connection using the Radio/Cable Patch Thru cable: AN/GRC-50 Patch Thru system 2 to TD-204 Patch Thru system 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

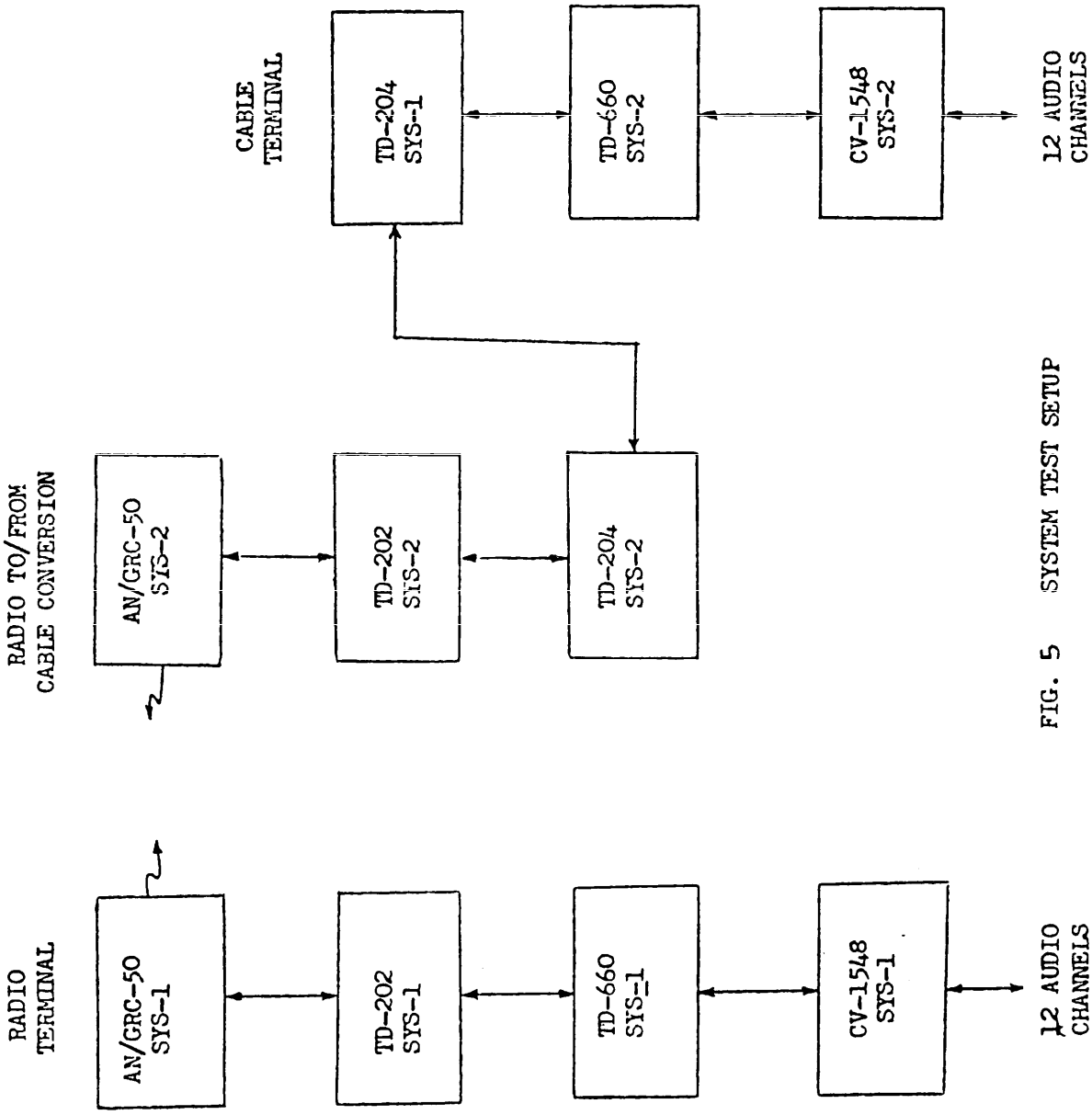


FIG. 5 SYSTEM TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

d. At Order Wire Patch Panel make the following cable connections using the Radio/Cable Order Wire patch cords: AN/GRC-50 Order Wire system 1 to TD-202 Order Wire system 1; and AN/GRC-50 Order Wire system 2 to TD-202 Order Wire system 2.

e. At Radio Patch Panel make the following cable connections using Radio Patch Cords:

AN/GRC-50 SYS-1	TD-202 SYS-1
RADIO IN	RADIO OUT
RADIO OUT	RADIO IN

AN/GRC-50 SYS-2	TD-202 SYS-2
RADIO IN	RADIO OUT
RADIO OUT	RADIO IN

f. At Cable Patch Panel make the following cable connections using Cable Patch cords:

Video ENT BOX SYS-1	TD-204 SYS-1
FROM CABLE	FROM CABLE
TO CABLE	TO CABLE

VIDEO ENT BOX SYS-2	TD-204 SYS-2
FROM CABLE	FROM CABLE
TO CABLE	TO CABLE

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

g. Interconnect system 1 and system 2 video connectors at Video Entrance Box using Cable Assembly CX-11230/G.

h. Repeat step 4.8.1.j.

i. Set channel switches of each CV-1548 to 2W.

j. Set 2 wire/4 wire switch of each TD-660 to 2 wire.

k. Place all channel switches at signal entrance panel to BINDING POST position.

l. Set Converters CV-1548, Multiplexers TD-660, Combiners TD-202 and TD-204, and Radio Sets AN/GRC-50 for proper operations in accordance with the provided instructions.

NOTE: The transmit and receive frequencies of the AN/GRC-50 must be separated by at least 15 channels.

m. Connect Handset H-156/U to each radio receiver; and Headset H-91A/U to each Combiner TD-204. Ring combiner 2, radio 2, and radio 1 from combiner 1; ring combiner 1, radio 1, and radio 2 from combiner 2; establish communications between all handsets and headsets.

n. Repeat steps 4.8.2k thru q.

4.9 Vehicular transportation test.- The facility shall be secured (and blocked to prevent shifting) to a government furnished 2½ ton, M-35 truck and transported over the specially prepared course described in 4.9.1 for 10 laps at a speed of 5 mph, 10 laps at a speed of 10 mph, 10 laps at a speed of 15 mph, and 10 laps at a speed of 18-20 mph. A lap is defined as traversing the course in one direction. The facility shall meet the requirements of paragraph 3.5 after completion of the test.

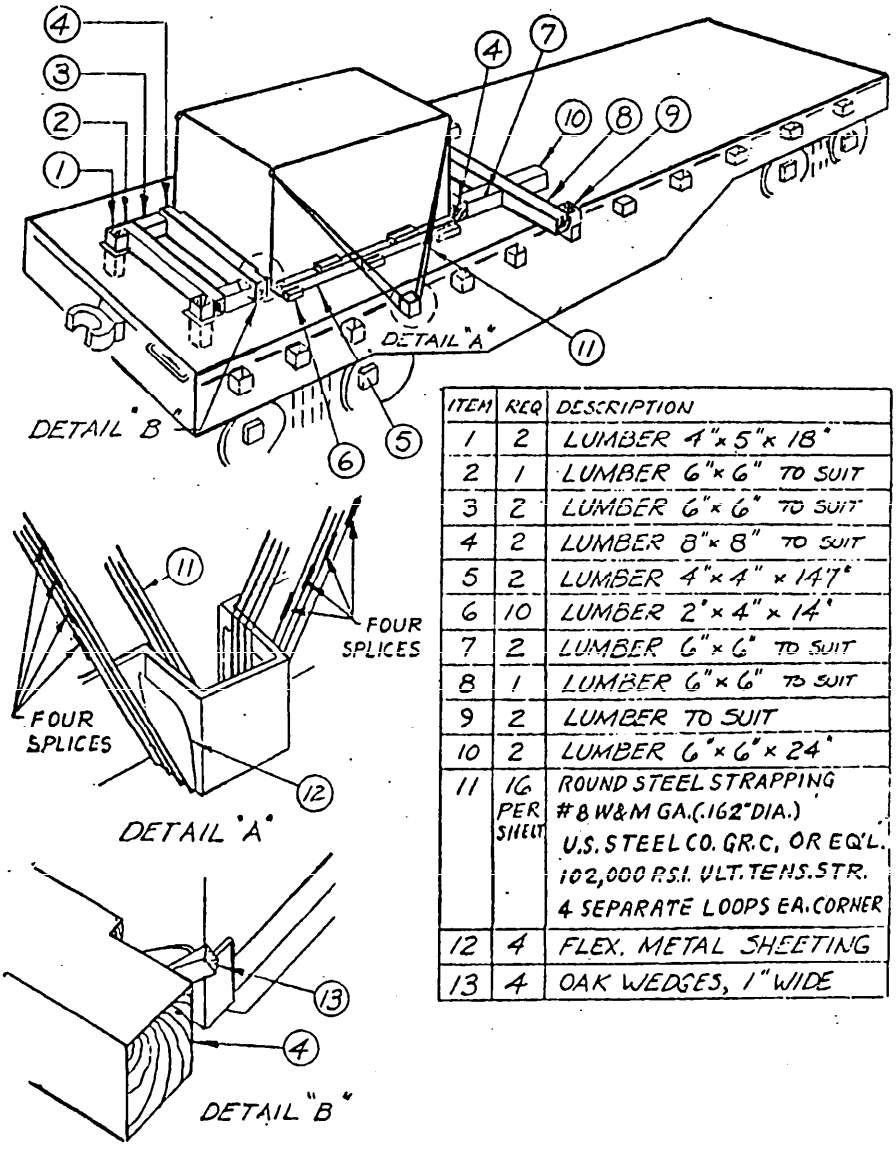
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

NOTE: The facility shall be loaded to simulate the AN/TRC-151()(V) using live and dummy loads. (See 6.2b for this test).

4.9.1 Vehicular transportation course.- The course for the military truck transportation shall have twelve (12) 4-inch by 8-inch timbers placed 25 feet apart on the 8-inch face and with the 4-inch face fully above ground. The top edges shall have a 1 x 1-inch chamfer. The sixth and twelfth timbers shall be placed at 45 degrees to the direction of travel, and all other timbers shall be placed perpendicular to the direction of travel. The timbers shall be anchored securely by any convenient means. The ground on which the timbers are placed shall be hard and level.

4.10 Rail transport, impact test.- This test shall be performed on a level stretch of track in the manner described in this paragraph. The test flatcar, timber blocking, and shelter facility tiedown strapping shall be as specified in Figure 6 and shall meet the provisions of the test as specified in 4.10.1; the test flatcar, less the shelter facility, and the two empty flatcars coupled to it shall each weigh 45,000 to 55,000 pounds. The loaded gondola car shall meet the provisions of the test as specified in 4.10.2 and shall have a total weight of 165,000 pounds. The loaded gondola car totaling 165,000 pounds traveling at 9 miles per hour (or equivalent energy conditions acceptable to the government) shall be impacted against the stationary loaded testcar coupled to the two empty flatcars, all with brakes off. This test shall be conducted four times; the first two runs shall be with the door-end of the shelter facility facing the loaded gondola car; the second two runs shall be with the door end of the shelter facility facing the opposite direction from the loaded gondola car. The lashing, blocking, and wedges holding the shelter facility in place shall be tight at the start of each run. If the blocking is torn loose or the lashing is broken during the impact, the run shall be repeated. After each impact, the shelter facility shall meet the requirements of 3.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.



ITEM	REQ	DESCRIPTION
1	2	LUMBER 4"x 5"x 18"
2	1	LUMBER 6"x 6" TO SUIT
3	2	LUMBER 6"x 6" TO SUIT
4	2	LUMBER 8"x 8" TO SUIT
5	2	LUMBER 4"x 4" x 147"
6	10	LUMBER 2"x 4"x 14"
7	2	LUMBER 6"x 6" TO SUIT
8	1	LUMBER 6"x 6" TO SUIT
9	2	LUMBER TO SUIT
10	2	LUMBER 6"x 6"x 24"
11	16 PER SHEET	ROUND STEEL STRAPPING #8 W&M GA.(.162"DIA.) U.S. STEEL CO. GR.C, OR EQ'L. 102,000 P.S.I. ULT. TENS. STR. 4 SEPARATE LOOPS EA. CORNER
12	4	FLEX. METAL SHEETING
13	4	OAK WEDGES, 1" WIDE

FIGURE 6 - SHELTER SECURED ON FLATCAR

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

4.10.1 Test flatcar.- The wooden planking of the test flatcar bed shall be in good condition. All 6 x 6 inch and 8 x 8 inch blocking shall be secured to the bed of the flatcar using 10 inch spikes on 12 inch centers. The 8 x 8 inch blocking shall be notched to provide ¼ inch clearance around the towing eye castings of the shelter; an oak wedge shall be installed into this clearance as shown in the figure referred to above at each of four corners of the shelter. The 4 x 4 inch and 2 x 4 inch blocking shall be nailed to the bed of the flatcar as shown in the figure referred to above. All blocking around the shelter facility shall be snugly against the shelter. (All other blocking shall be snugly abutting as shown in the figure.) The tiedown strapping shall be installed using a strapping machine U.S. Steel Corp., Model GT, or equal. If any tiedown strapping loosens or breaks during any run, it shall be replaced before the next run using new wire and a strapping machine as specified above.

4.10.2 Loaded gondola car.- The load in the loaded gondola car shall be one of the following:

a. A coarse size of either coal, crushed stone, gravel, crushed concrete, metal scrap other than sheet metal or metal shavings or other springy forms of metal.

b. Chunks, slab, block, or plate form of the preceding acceptable materials, constrained to prevent shifting during impact.

4.11 Drop tests.-

4.11.1 Flat drop.- The shelter facility loaded as in 4.9 shall be lifted 18 inches from the ground and allowed to fall freely with the skids impacting onto a hard substantial concrete surface. The test shall be performed once. Upon completion of this drop, the facility shall meet the requirements of 3.7.

4.11.2 Rotational drop.- The shelter facility loaded as in 4.9 shall have one skid resting on a nominal 4 inch high timber. The outer edge of the opposite skid shall be raised 18 inches and the shelter facility shall be allowed to fall freely onto a hard substantial concrete surface. This impact shall be performed four (4) times, once for each bottom edge of the shelter facility. On completion of these drops, the shelter facility shall meet the requirements of 3.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

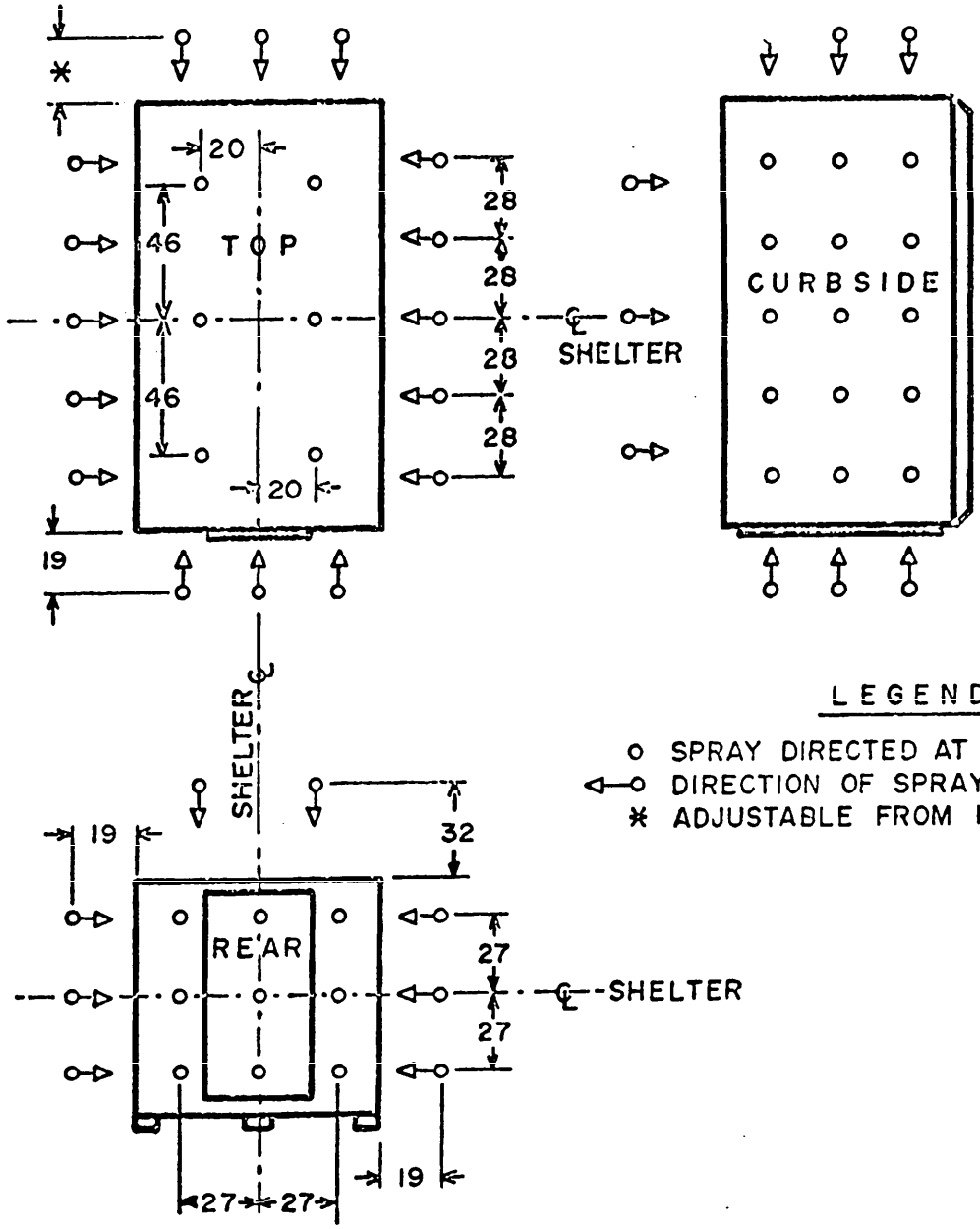
4.12 Watertightness test.- The shelter facility,without the use of any additional external sealing, caulking, taping, etc. shall be subjected to the following two part test: (Note. Perform tests with inside covers of entrance boxes removed).

a. Each of the five exposed surfaces of the shelter facility, with covers closed, shall be sprayed with water from nozzles, model G29SQ (or GG29SQ) as made by Spraying Systems Company, Bellwood,Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using fifteen nozzles, and the roof using six nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches. All of the five surfaces shall be simultaneously subjected to the test of this paragraph unless otherwise specified by the contracting officer. (See Figure 7 for nozzle location). Each panel shall be sprayed for a period of 20 minutes. (If the watertightness test is to be followed by the fording test of 4.13, the bottom row of nozzles may be eliminated with the exception of the nozzles facing the door-end panel). Upon completion of the above test, the shelter facility shall meet the requirements of 3.8. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

b. Open all covers to operating position (with exception of the door vent panel at the rear of the shelter) and repeat a above.

4.13 Fording test.- The shelter facility shall be immersed in water for 1 hour to a depth of 21 inches (measured from the bottom of the skids) and meet the requirements of 3.9. No special fording kits shall be used. The shelter shall be restrained from floating by external tiedowns. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.



NOTES

1 NOZZLES SHALL BE PERPENDICULAR TO SHELTER.
2 UNLESS OTHERWISE SHOWN ALL DIMENSIONS SHALL BE TAKEN FROM INDICATED CENTER LINES.
3 GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES.
4 ROADSIDE NOZZLE LOCATIONS SAME AS CURBSIDE.
5 FRONT NOZZLE LOCATIONS SAME AS REAR.
6 DIMENSIONS ARE IN INCHES.

NOZZLE LOCATIONS

FIGURE 7

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

4.14 Mounting insert tests.-

4.14.1 Preproduction and Group C insert test.- Four of each size insert shall be randomly selected for test. Each selected insert shall be tested with an axial load followed by a torque load in accordance with Table V. After test, the requirements of 3.10 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

4.14.2 Production Group B insert test.- Five of each size insert (if the total quantity of any size is less than five, all of that size shall be tested) shall be randomly selected for test in each sample shelter facility. Each selected insert shall be tested in accordance with Table V. After test, the requirements of 3.10 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

Table V - Mounting inserts loads

Insert	Group C and Preproduction Test		Group B Production Test	
	Axial(lbs)	Torque(in-lbs)	(Axial(lbs)	Torque(in-lbs)
5/16-18	2000	100	1000	100
1/4-20	1300	60	650	60
#10-32	600	23	300	23

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

4.15 Testing of solderless terminal-lug connections.- Connections made with solderless terminal-lugs shall be tested for conformance to 3.20.4.

4.16 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.21. When a dimension is not within specified or design limits, it shall be considered a defect.

a. Size and mounting dimensions of brackets and mounting plates.

b. Size and mounting dimensions of racks, patch panels, and cabinets.

4.17 Visual and mechanical inspection.- The shelter facility shall be examined for the defects listed in Standard MIL-STD-252, the requirements of 3.22 and Table VI.

Table VI - Classification of Visual and Mechanical Defects

Classification	Defects
Major	1. Mounting plates missing or mislocated. 2. Aircraft loading data plate, instruction or nameplate omitted, incorrect or illegible. 3. Welds - cracked or porous. 4. Sealer improperly applied (holes, separations, or lack of adhesion). 5. Mounting inserts missing or mislocated. 6. Burrs or sharp edges in wire duct not removed. 7. Equipments or bracketry loose or missing. 8. Doors, etc. inoperative. 9. Electrical receptacles, switches, jacks, connections, wire ducts or other electrical components loose, improperly located or inoperative. 10. Locking or holding devices missing or inoperative. 11. Gaps between sections of wire duct exceed 1/16".

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

Table VI - Classification of Visual and Mechanical Defects(cont'd)

Classification	Defects
Minor	1. Doors not easily operable. 2. Fastening devices difficult to operate. 3. Finish - abrasions or scratches.

4.18 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.19 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by rough handling:

Paragraph 4.7 Electrical installation

Paragraph 4.17 Visual and mechanical

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation, packaging, and packing of the shelter facility shall be in accordance with the applicable requirements of Specification MIL-S-55505.

6. NOTES

6.1 Intended use.- The Shelter,Electrical Equipment S-330()/TRC-117()(V) is the shelter facility portion of the AN/TRC-151()(V) and it is used to house the operational equipments including:

a. 2 ea AN/GRC-50 Radio Set

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

b. 2 ea CV-1548()/U Converter,Telephone Signal

c. 2 ea TD-660()/G Multiplexer

d. 2 ea TD-202()/U Multiplexer

e. 2 ea TD-204()/G Multiplexer or TD-754()/G Multiplexer

f. 1 ea CN-514()/GRC Voltage Regulator

g. 1 ea TA-312()/PT Telephone

h. 1 ea LS-147()/FI Intercom

forming the Radio Terminal Set.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Preproduction and Group C inspection of the shelter facility requires use of contractor fabricated dummy loads for the government furnished live loads (radio components) unless otherwise specified in the bid request and contract.

c. A list of government furnished equipments required as part of the S-330()/TRC-117(V).

d. A list of equipments to be government loaned for testing purposes.

e. Submission of the statement referenced in 3.18.4 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

6.3.1 Examination.- Examination consists of simple, generally non-destructive determinations of compliance, without use of special testing equipment.

6.3.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the government, pending the decision from the contracting officer on the adequacy of corrective action.(See 4.5.3.4)

6.5 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: S-330R/TRC-117(V). The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.6 Verification inspection.- Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
Army (EL)

Preparing Activity:
Army - EL

Project No. 5820-A494

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the current version before use.

To detach this form, cut along this line

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-S-55557A(EL), SHELTER, ELECTRICAL EQUIPMENT S-330( )/TRC-117(V)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding General
U.S.Army Electronics Command
ATTN: AMSEL-~~TD~~-SS
Fort Monmouth, New Jersey 07703

To detach this form, cut along this line

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.